There are two APIs using which games are developed. The first is DirectX, and the second, OpenGL

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Graphics card drivers are software that help the card work seamlessly with the rest of the system, and can enable better performance. Graphics card vendors usually release updated versions of these drivers on a regular basis, so you can get that little extra from your card.

Many features and enhancements depend on the way drivers are written. Some games are specifically optimised for certain versions of drivers, and vice-versa. Here are some critical features to get more performance from your games.

There are two APIs (Application Program Interface) using which games are developed. The first is Microsoft's DirectX, of which Direct3D is a critical component. The second is OpenGL—Open Game Language.

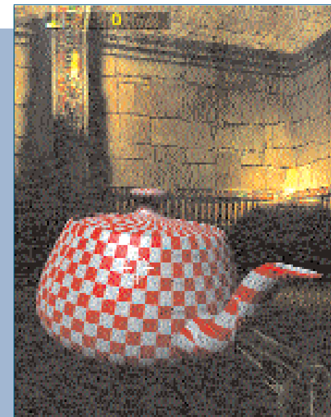
Games developed using either of these APIs use different sets of features, which can be enabled or disabled to improve quality and/or performance. We've classified the various features under 'Direct3D' and 'OpenGL'.

DIRECT3D

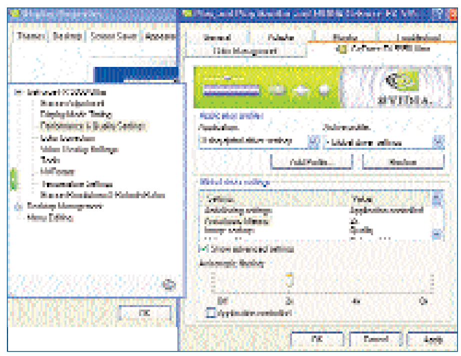
Bump Mapping

This feature—when enabled—allows textures to show depth. Tiles on the floor, or a tiled stone wall, will actually appear bumpy. This feature was introduced in DirectX 6 and thus, most current-generation games support this feature by default.

When enabled for older games, it will improve image quality, but reduce performance. This, again, depends on the game, and whether the game supports this feature. Games these days use two types of bump mapping—EMBM (environment mapped bump mapping), and DOT3 bump mapping.



This image shows how bump mapping is used to create the emboss effect on the kettle



Adust the Anisotropic Filtering in the nVidia control panel

EMBM can generate the bump mapping effect with intricate details such as the pock-marked surface of bricks, scratches on a metallic surface, realistic reflections on water ripples, and more. This was introduced in the Matrox G400 graphics chip for the first time back in 1999.

DOT3 bump mapping, also known as

Graphics Jargon

Anti-Aliasing

This is a technique that removes jaggedness at the edges of an object. Since it eats up lots of GPU cycles, a higher setting will lower the frames per second that you can get. Most cards are set by default to the lowest setting of 2x, and the 'Application Preference' option lets you define the setting from the game.

Anisotropic Filtering

This filtering technique—when enabled—lets you see distant objects clearly. A good example is a straight road. With this setting enabled, you will be able to see the texture even at the farthest point clearly. Similar to Anti-Aliasing, the lowest setting is 2x, and there is also the 'Application Preference' option. A higher setting of, say, 8x, will dramatically improve image quality, but will also adversely affect the output frames per second.

Texture Compression

This is a technique, using which, a compression algorithm built into the hardware of the graphics processor compresses and decompresses the texture data, which results in an increase in the effective bandwidth of the graphics data path between the graphics chip and the CPU.

This was developed by S3 for OpenGL applications, and was known as S3TC. It was later adopted by Microsoft and used in DirectX, and became known as DXTC (DirectX Texture Compression). This feature requires that the hardware be capable of supporting it.